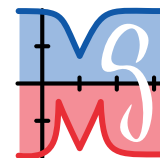


MoleGraph: The Open Gateway to Science

Empower your **STEM** education and Inquiry-Based Learning (IBL) with MoleGraph. This versatile **open-source DIY Probeware system**, powered by **Arduino NANO**, invites you to look inside the "black box" of scientific measurement. Build your **own instruments**, analyze real-time data via Bluetooth, and explore the principles of electronics. Designed for limitless creativity, the system is fully compatible with the **m-BITBEAM** construction system, allowing you to easily integrate sensors into custom experiments and robotic projects.



Group name

List of sensors in this box (checkmarks)

ID	Name	Measured Quantities [Unit]
1	Thermometer (DS18B20)	Temperature [°C] <i>Measuring range -55..+125 °C. Output Resolution: 9-bit to 12-bit (programmable), accuracy: ±0.5 °C. Conversion time: 750 ms at 12-bit.</i>
2	IR Thermometer (MLX90614)	Temperature [°C], Int. Temperature [°C] <i>Measuring range: -40..+125 °C for sensor temperature and -70..+380 °C for object temperature. Resolution 0.02 °C, accuracy ±0.5 °C. Maximum sampling rate 1000 Hz.</i>
3	Pressure sensor (MPX5700DP)	Pressure [Pa] <i>Measuring range: 0..750 kPa. Resolution 0.7 kPa (10-bit), accuracy ±2.5 %. Maximum sampling rate 1000 Hz. Operating Temperature: -40 to 125 °C.</i>
4	pH sensor (E201 + pH-4502C)	pH [], Voltage [V], RAW [], Trigger [] <i>Measuring range 0..14 pH. Resolution 0.02 pH (10-bit), accuracy: ±0.1. Analog output - calibration is needed!</i>
5	Conductivity sensor (TDS)	Conductivity [µS/cm], Salinity (ppm) [ppm], RAW [], Voltage [V], Trigger [] <i>Conductivity/TDS - Total Dissolved Solids. Measuring range: conductivity 0..1650 µS/cm, salinity (TDS) 0..1000 ppm. Resolution 2 µS/cm (1 ppm) (10bit), accuracy: ±5 %. Analog output - calibration is needed!</i>
6	Light sensor (photoresistor module)	Illuminance [RAW], Voltage [V], Trigger [] <i>Measuring range 0..5 V (10-bit converter) – output only as RAW value 0..1023, voltage 0..5 or trigger 0/1.</i>
7	Light sensor (BH1750 luxmeter)	Illuminance [lux] <i>Measuring range 0..65 535 lux (16-bit converter). Resolution 1 lux, accuracy 1 lux. Measurement time is approx 120 ms (max sampling 8 Hz).</i>
8	Alcohol gas sensor (MQ-3) - ethanol, gasoline	Concentration (ppm) [ppm], Concentration (%) [%], Voltage [V], RAW [], Trigger [] <i>Measuring range 25..500 ppm. Resolution 2.5 ppm (10-bit), accuracy around ±10 ppm. Analog output - calibration is needed!</i>
9	Combustible Gas (MQ-2) - propane, methane, butane	Concentration [ppm], Concentration [%], Voltage [V], RAW [], Trigger [] <i>Measuring range 0..10 000 ppm. Resolution 10 ppm (10-bit), accuracy around ±15 %. Analog output - calibration is needed!</i>
10	CO ₂ gas sensor (MH-Z16)	Concentration [ppm], Concentration [%] <i>Measuring range 0..50 000 ppm. Resolution 0.05 ppm, accuracy around ±50 ppm. Maximum sampling rate 0.5 Hz.</i>
11	Magnetic field sensor (Hall module KY-024)	RAW [], Voltage [V], Trigger [] <i>Measuring range 0..5 V (10-bit converter) – output only as RAW value 0..1023, voltage 0..5 or trigger 0/1.</i>
12	Sound sensor (microphone KY-037)	RAW [], Voltage [V], Trigger [] <i>Measuring range 0..5 V (10-bit converter) – output only as RAW value 0..1023, voltage 0..5 or trigger 0/1.</i>
13	DC Voltage sensor 0-25 V (voltage divider)	Voltage [V], RAW [] <i>Measuring range 0..25 V. Resolution 25 mV (10-bit), accuracy around ±25 mV.</i>
14	AC/DC Current sensor 0-5 A (ACS712)	Current [A], RAW [] <i>Measuring range 0..±5 A. Resolution 10 mA (10-bit), accuracy around ±1.5 %.</i>
15	Motion/distance US sensor (SR-HC04)	Distance [m], Velocity [m/s], Acceleration [m/s ²] <i>Measuring range 2..400 cm, measuring angle 15°. Accuracy ±3 mm. The Module automatically sends eight 40 kHz and detect whether there is a pulse signal back. Suggest to use over 60 ms measurement cycle, in order to prevent trigger signal to the echo signal.</i>

<http://www.e-mole.cz/diy/molegraph>
<http://www.e-mole.cz/diy/molegraph-shield>

Applications for platforms: MS Windows, macOS, Linux, Android

Download on GitHub



Designed & Coded in Czech Republic

MoleGraph v4.3

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